Pseudodifferential Operators

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1 Differential Operators

We would like to generalize the notion of a differential operator on $C^\infty(R^n)$ to sections of vector bundles on a manifold. Consider the following system of equations:

$$\begin{aligned} 3\tfrac{\partial}{\partial x}f(x,y,z)+\tfrac{\partial}{\partial x}g(x,y,z)+2\tfrac{\partial}{\partial z}h(x,y,z) &= 0 \\ 3\tfrac{\partial}{\partial x}f(x,y,z)+3\tfrac{\partial}{\partial x}f(x,y,z)+3f(x,y,z) &= 0 \end{aligned}$$

We can represent the right side as an operator D applied to an R^3 valued function, where

$$D = \begin{bmatrix} 3 & 1 & 0 \\ 3 & 1 & 0 \end{bmatrix}\frac{\partial}{\partial x} + \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}\frac{\partial}{\partial z} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

In other words, D is a linear map $C^\infty(\mathbb{R}^2,\mathbb{R}^3) \to C^\infty(\mathbb{R}^2,\mathbb{R}^2)$. This suggests that instead of considering differential operators acting on functions (sections of a trivial line bundle,) we consider differential operators which act on sections of vector bundles.

In the following section, we let X be a manifold, with E and F smooth complex vector bundles on X. We denote by $\Gamma(E)$ the space of smooth C^∞ sections of E.

Definition 1.1. A differential operator of order m on X is a linear map $P : \Gamma(E) \to \Gamma(F)$, such that if we choose local coordinates x_i in an open set $U \subset X$, and also local trivializations $E|_U \simeq U \times \mathbb{C}^p$ and $F|_U \simeq U \times \mathbb{C}^q$, then P has the form

$$P = \sum_{|\alpha|\le m} A^\alpha(x)\frac{\partial^{|\alpha|}}{\partial x^\alpha}$$

where the A^α's are smooth $p \times q$ matrices of smooth functions, and such that one of the highest order coefficients is nonzero somewhere.

Observe that if we make a change of local trivializations of $E|_U$ and $F|_U$ by maps $g_E : U \to GL_p(\mathbb{C})$ and $g_F : U \to GL_p(\mathbb{C})$, then in these trivializations, P has the form

$$P = g_F\left(\sum_{|\alpha|\le m} A^\alpha \frac{\partial^{|\alpha}}{\partial x^\alpha}\right)g_E^{-1} = \sum_{|\alpha|\le m}\tilde{A}^\alpha\frac{\partial^{|\alpha|}}{\partial x^\alpha},$$

where the coefficients $\tilde{A}^\alpha$ can be obtained using the product rule to interchange the orders of ∂_x and g_E^{-1}. Namely, for all $|\alpha| = m$, we have the formula

$$\tilde{A}^\alpha = g_E A^\alpha g_E^{-1}$$

for the modified coefficients.

Similarly, if we change to coordinates $\tilde{x} = \tilde{x}(x)$ on the set U, P takes the form

$$P = \sum_{|\alpha| \le m} \tilde{A}^\alpha(\tilde{x}) \frac{\partial^{|\alpha|}}{\partial x^\alpha}.$$

Using the Jacobian for the change of variables, for any j we can write

$$\frac{\partial}{\partial x_j} = \sum_{k=1}^n \frac{\partial \tilde{x}_k}{\partial x_j} \frac{\partial}{\partial \tilde{x}_k}.$$

For any multi-index α with $|\alpha| = m$, we can write

$$\frac{\partial}{\partial x^\alpha} = \sum_{|\beta|=m} \left[\frac{\partial \tilde{x}}{\partial x}\right]_\beta^\alpha \frac{\partial}{\partial \tilde{x}^\beta},$$

where $\left[\frac{\partial \tilde{x}}{\partial x}\right]_*^*$ denotes the symmetrization of the m-th exterior power of the jacobian. Note that both upper and lower indices are symmetrized to reflect the commutativity of partial derivatives.

Thus we have

$$\tilde{A}^\alpha = \sum_{|\beta|=m} A^\beta \left[\frac{\partial \tilde{x}}{\partial x}\right]_\beta^\alpha$$

for the top order coefficients with $|\alpha| = m$.

We see that P behaves well under coordinate changes and changes of trivialization, so long as we only consider the top order terms. In fact, we have proven that the coefficients $\{i^m A^\alpha\}_{|\alpha|=m}$ represent a well defined section $\sigma(P)$ of the bundle $(\odot^m TX) \otimes \mathcal{H}om(E, F)$, where $\odot$ denotes the symmetric tensor product.

Next we note that the space $\odot^m V$ is canonically isomorphic to the space of homogeneous polynomial functions of degree m on V^*. Hence, for each cotangent vector $\xi \in T_x^* X$, the principal symbol gives a linear map

$$\sigma_\xi(P) : E_x \to F_x.$$

In other words, the symbol can be considered as a (polynomial) function on the cotangent bundle.

1.1 Examples on Riemannian Manifolds

Now let X be a riemannian manifold equipped with a metric $\langle \cdot, \cdot \rangle$. In local coordinates we can express the metric using the basis as

$$\sum g_{jk} dx_j dx_k$$

for some matrix g_{jk}. We also denote by g^{jk} the inverse of g_{jk}, and $g := \det(\{g_{jk}\})$.

Example. Consider the Laplace-Beltrami operator $\Delta : C^\infty(X) \to C^\infty(X)$ defined by $\Delta = d^* d$, where d^* is defined on k-forms α by $d^*\alpha = (-1)^{(nk+n+1)} * d^* \alpha$. We first calculate an explicit formula for d^* applied to k-forms $\alpha = \sum A_i dx^i$:

$$d^*\alpha = \frac{1}{\sqrt{g}} \sum_{ij} \frac{\partial}{\partial x_j}(A_i g^{ij} \sqrt{g}).$$

Then we can apply it to $\alpha = df = \sum \frac{\partial f}{\partial x_i} dx^i$ to obtain the following formula for Δ:

$$\Delta f = \frac{1}{\sqrt{g}} \sum_{ij} \frac{\partial}{\partial x_j} (\sqrt{g} g^{ij} \frac{\partial f}{\partial x_i}) \tag{1}$$

$$= \sum_{j,k=1}^{n} g^{jk} \frac{\partial^2 f}{\partial x_j \partial x_k} + \{\text{lower order terms}\}. \tag{2}$$

Now we can calculate its symbol: for some cotangent vector $\xi = \sum \xi_k dx^k$, the polynomial function is given by

$$\sigma_\xi(\Delta) = i^2 \sum_{j,k=1}^{n} g^{jk} \xi_j \xi_k = -\|\xi\|^2.$$

Example. From this discussion, it is natural to wonder if we can just define an operator with symbol $i\|\xi\|$. In fact we could, but it will be pseudodifferential and not quite what we are after in several ways.

Instead, we introduce the concept of clifford action of T^*M on a bundle S. In defining the dirac operator D in the usual way, we see that its symbol is

$$\sigma_\xi(D) = ic(\xi),$$

where $c(\xi) \in \text{End}, mk(S)$ denotes clifford multiplication by ξ. Trivializing S in some neighborhood U, $c(\xi)$ becomes some matrix-valued function on U, and thus D is an ordinary differential operator on S.

Remark. For a differential operator $P : \Sigma(E) \to \Sigma(F)$ we can consider $\sigma(P)$ as a bundle map of the pullbacks $\pi^*E \to \pi^*F$ in the natural way, under the projection $\pi : T^*M \to M$.

2 Sobolev Spaces

I don't want to spend too much time talking about these, but the results are important for the next section.

The point of this section is to introduce the fourier transform-based sobolev spaces H^s for $s \in \mathbb{R}$, and to extend them to sections of vector bundles in a suitable way.

Definition 2.1. Let E be a hermitian vector bundle with connection ∇ on a compact Riemannian manifold X. Given $u \in \Sigma(E)$, we have $\nabla u \in \sigma(T^*X \otimes E)$. Furthermore, using the tensor product connection on $T^*X \otimes E$, $\nabla\nabla u \in \sigma(T^*X \otimes T^*X \otimes E)$. Thus for any k, we can define a norm

$$\|u\|_k^2 = \sum_{j=0}^{k} \int_X |\nabla^k u|,$$

called the basic Sobolev k-norm on $\sigma(E)$.

The completion of $\sigma(E)$ under this norm is the Sobolev space $H^k(E)$.

A fundamental fact about sobolev spaces is the following proposition:

Proposition 2.1. A differential operator $P : \sigma(E) \to \sigma(E)$ of order m is bounded as a linear map $P : H^k \to H^{k-m}$. Hence, P extends to a linear map $P : H^k \to H^{k-m}$.

The next construction reduces the study of sections of E to the study of smooth $\mathbb{C}^p$-valued functions with compact support in the unit ball using a partition of unity. Then we can build a norm on $\sigma(E)$ using the fourier transform on sections in each trivialized patch of E under the transformed coordinates. I will not provide it here, but we need an open cover of X consisting of trivialized patches of E, and a partition of unity subordinate to this cover, such that the open cover is sufficiently overlapping. Then we change coordinates with a bijection $[-1, 1] \to \mathbb{R}$.

2.1 Fourier transform-based Sobolev spaces

Here we briefly outline the sobolev spaces $H^s(\mathbb{R}^n)$, with values in $\mathbb{C}^p$, for $s \in \mathbb{R}$. These are constructed using the fourier transform.

Definition 2.2. The Fourier Transform $\hat{u} : \mathbb{R}^n \to \mathbb{C}^n$ of a compactly supported function $u \in C^\infty(\mathbb{R}^n, \mathbb{C}^n)$ is defined as

$$\hat{u}(\xi) = (2\pi)^{-n/2} \int_{\mathbb{R}^n} e^{-i\langle x,\xi\rangle} u(x) dx.$$

Denote the Fourier map by $\mathcal{F}$.

In fact, $\mathcal{F}$ is an isometry $\mathscr{S} \to \mathscr{S}$, where $\mathscr{S}$ is the space of Schwartz functions of "rapidly decreasing functions," who are dominated by polynomials of arbitrarily large negative order on any compact set, under the L^2 norm.

Definition 2.3. We define the Sobolev s-norm $\|\cdot\|_s$ by the formula

$$\|u\|_s^2 = \int (1 + \|\xi\|)^{2s} |\hat{u}(\xi)|^2 d\xi.$$

To motivate this definition, note that for compactly supported u on $\mathbb{R}$, after integrating by parts, we have

$$\widehat{(\partial_x u)} = \int e^{ix\xi} \partial_x(u)(x) dx = \int \xi e^{ix\xi} u(x) dx = \xi \hat{u}.$$

Thus for the case $s = 1$, we have

$$\int |\xi|^2 |\hat{u}(\xi)|^2 d\xi = \int |\xi \hat{u}(\xi)|^2 d\xi = \int |\widehat{(\partial_x u)}(\xi)|^2 d\xi = \int |\partial_x u(x)|^2 dx$$

since $\mathcal{F}$ is an isometry in the L^2 norm.

One surprising result is the following:

Proposition 2.2 (The Sobolev Embedding Theorem)**.** Fix a positive integer k. Then for each real number $s > (n/2) + k$, there is a constant K_s such that for every $u \in \mathscr{S}$,

$$\|u\|_{C^k} \le K_s \|u\|_s.$$

Hence, there is a continuous embedding

$$H^s \subset C^k.$$

In order to extend these results globally to define Sobolev spaces on compact manifolds, we need a couple of lemmas, which we will not prove, but are not too unreasonable.

Lemma 2.1. Let A be a smooth matrix-valued function on $\mathbb{R}^n$ so that $|D^\alpha A|$ is bounded for all α. Then the map $T : \mathscr{S} \to \mathscr{S}$ given by $Tu = Au$ extends to a bounded linear map $T : H^s \to H^s$ for all $s \in \mathbb{R}$.

Lemma 2.2. Let $\Phi : \mathbb{R}^n \to \mathbb{R}^n$ be a C^∞ diffeomorphism which is linear outside a compact set. Then the map $T : \mathscr{S} \to \mathscr{S}$ given by $Tu = u \circ \Phi$ extends to a bounded linear map $T : H^s \to H^s$ for all $s \in \mathbb{R}$.

Now we globalize. Let E be a smooth vector bundle over a compact manifold X. Fix a good presentation $\{U_\beta\}$ for E with coordinates $x_\beta : U_\beta \to \mathbb{R}^n$ and a partition of unity $\{\chi_\beta\}$ subordinate to the cover $\{B_\beta\} = \{x_\beta^{-1}(B^n)\}$.

Since $\sum \chi_\beta = 1$, any $u \in \Gamma(E)$ can be written $u = \sum u_\beta$ where $u_\beta = \chi_\beta u$. For $s \in \mathbb{R}$, define a Sobolev s-norm on $\Gamma(E)$ by setting

$$\|u\|_s = \sum_{\beta=1}^{N} \|u_\beta\|_s,$$

where the $\|u_\beta\|_s$ are defined in terms of the presentation $E|_{U_\beta} \simeq \mathbb{R}^n \times \mathbb{C}^n$.

As a consequence of 2.1 and 2.2, we have the following:

Proposition 2.3. For any $s \in \mathbb{R}$ and any smooth vector bundle E over a compact manifold, the equivalence class of the norm $\|\cdot\|_s$ as defined above is independent of the good presentation used to define it. Furthermore, when $s = k$ is an integer, the equivalence class of $\|\cdot\|_k$ is the same as that defined in the first definition 2.

Moreover, we have an analagous result to 2.1 for differential operators:

Proposition 2.4. Let $P : \Gamma(E) \to \Gamma(F)$ be a differential operator of order m as defined in whose (locally expressed) coefficients are bounded as in 2.1. Then $P : \Gamma(E) \to \Gamma(E)$ extends to a bounded linear map $P : H^s(E) \to H^{s-m}(F)$ for all $s \in \mathbb{R}$.

Remark. In fact, $H^s(E)$ is a hilbert space under the H^s inner product

$$\langle u, v\rangle_s = \int (1 + |\xi|)^{2s} \hat{u}(\xi)\overline{\hat{v}(\xi)} d\xi.$$

Alternatively, we can define a pairing $\langle \cdot, \cdot\rangle_* : H^s(E) \times H^{-s}(E^*) \to \mathbb{R}$ by setting

$$\langle u, v\rangle_* = \int u(v),$$

placing $H^s(E)$ and $H^{-s}(E^*)$ in duality.